

PROJECT RESEARCH DOCUMENT

Improving Trust and Accessibility in Student Housing: The Student Finder Accommodation (SFA) Web Platform

Student Number: 261337665

1. Research Title

Improving Trust and Accessibility in Student Housing: The Student Finder Accommodation (SFA) Web Platform

This title identifies the core problem being addressed – the difficulty all South African students face in verifying safe, legitimate, and suitably funded accommodation, whether they are self-funded, bursary-funded, or NSFAS-funded – and names the proposed solution: SFA, a web-based student accommodation finder built around landlord verification, scam prevention, flexible funding/price filtering, and AI-assisted support for every student, not only those on NSFAS.

2. Aim & Objectives

Aim

To design and develop a web-based student accommodation platform (SFA) that helps all South African students – whether self-funded, bursary-funded, or NSFAS-funded – find safe, verified, and affordable housing more efficiently than existing platforms, by combining landlord verification, scam detection, flexible funding/price filtering (including but not limited to NSFAS accreditation), and AI-assisted support.

Objectives

1. To research and evaluate existing student accommodation platforms, including ResiFind, in order to identify gaps in verification, funding/price filtering, and student support features that affect the broader student population.
2. To design a user-friendly front-end interface (landing/browse page, listing-detail page, booking flow, and login/signup page) that allows any student to filter accommodation by price, funding type (including an NSFAS option), and verification status.
3. To plan the system architecture – including IPO charts, flowcharts, pseudocode, and UML use case diagrams – that supports listing verification, scam detection, and a secure booking process for all users.
4. To develop an AI chatbot, StudentStay, that guides students through accommodation queries, funding and eligibility questions (including but not limited to NSFAS), and the application process.
5. To test and evaluate the prototype's usability and functionality against the gap identified in the chosen existing solution, across different student funding profiles.

3. Progress Report

3.1 Research completed

A competitor scan was carried out across five existing student accommodation platforms operating in South Africa: ONSA, Dwell, ResiFind, OffCamp, and UniHub. For each, we reviewed core features, pricing model, target market, and stated safety/verification measures. From this scan, two unique selling points were identified for SFA relative to all five competitors: an NSFAS-accreditation filter (offered alongside, not instead of, general price and funding filters) and verified landlord trust badges, both aimed at benefiting the full student population rather than one funding group. Supporting research included a short statistics brief on the South African student housing market, a business plan outlining the platform's value proposition and revenue model, and financial computations to support the business case.

3.2 Design work completed

System design has been documented through IPO (Input-Process-Output) charts and system flowcharts covering the core user journeys (searching for accommodation, filtering by price and funding type, booking, and landlord verification), along with pseudocode for the key logic and UML use case diagrams showing student, landlord, and administrator interactions. A visual identity was also finalised: a colour palette of Deep Forest Green, Fresh Green, Mint/Sage, Warm Sand, Verified Blue, Alert Red, and NSFAS Gold, documented in a colour/style guide, together with branded slide decks used for presentations.

3.3 Coding completed

A front-end HTML prototype has been built and is functional, using the platform's brand palette and localised to South African Rand pricing. Completed pages include:

- A browse/landing page with live listing filters, including price, location, and an optional NSFAS accreditation toggle for students who need it.
- A listing-detail page with a booking form and contact/report-listing buttons.
- A login/signup page with tabbed forms and a student/landlord role toggle.
- A shared, responsive mobile hamburger menu used consistently across all pages.

3.4 Outstanding work

The following items are still to be completed before the prototype is submission-ready:

- Back-end and database integration so listings, bookings, and user accounts persist beyond the front-end mock-up.
- Functional build-out of the StudentStay AI chatbot (currently planned/designed but not yet coded).
- Implementation of the scam-detection logic referenced in the design documentation.
- User/usability testing of the prototype and incorporation of feedback.
- Finalising the written project report and rehearsing the project presentation/demo.

4. Problem Statement

4.1 What the existing business (ResiFind) does

ResiFind is a real, operating South African student accommodation platform (app and website) that markets itself as a “safety-first” solution. It allows students to search verified listings near their university by location, price, and amenities, with map-based search and distance-to-campus information. Students can apply digitally, uploading documents and tracking their application status in real time, and can message landlords

directly inside the app rather than sharing personal contact details. It also includes a safety feature in the form of an SOS emergency alert system.

4.2 The gap in ResiFind's current solution

ResiFind's marketing and feature set focus on general listing verification, safety, and application tracking, but its filtering does not clearly separate different funding situations – there is no dedicated way for a student to filter specifically for NSFAS-accredited residences, nor is there any tailored guidance for students on bursaries or other funding arrangements who have their own eligibility questions. It also offers no AI-assisted guidance of any kind: whether a student's question is about NSFAS accreditation, a bursary requirement, or simply understanding a listing's terms, they have to rely on manually messaging a landlord or searching outside the app for answers. This leaves a gap for the broader student population, not just NSFAS-funded students, who all need quick, reliable answers about funding fit and listing legitimacy.

4.3 How the SFA prototype addresses this gap

SFA is being built to close this broader gap rather than compete on general listing volume. Where ResiFind treats verification as a single, general safety feature, SFA lets any student filter accommodation by price and funding type, with NSFAS accreditation offered as one option among several rather than the only lens on the platform, alongside a separate verified-landlord trust badge so funding fit and landlord legitimacy are visible independently. SFA is also introducing the StudentStay AI chatbot to answer accommodation, funding, and application questions instantly for every student, reducing reliance on back-and-forth messaging with landlords regardless of how a student's studies are funded. Combined with the scam-detection feature planned in the system design, SFA's approach targets the points – funding clarity and guided support for all students – that ResiFind's current offering does not cover.